

CUSTOMER CHURN PREDICTION SYSTEM

Complete Analytical Report

Exploratory Analysis • Statistical Testing • Machine Learning • Explainability

PROJECT AT A GLANCE	DETAIL
Dataset	IBM Telco Customer Churn
Records / raw variables	7,043 customers / 21 columns
Modeling split	5,634 training / 1,409 test; stratified
Final candidate	Balanced tuned XGBoost
Operating threshold	0.45, selected using training OOF predictions

Executive Summary

This project builds an end-to-end workflow to identify customers at risk of churn and explain model predictions. The notebook covers data inspection and cleaning, exploratory analysis, statistical tests, feature engineering, preprocessing, baseline benchmarking, imbalance handling, hyperparameter tuning, threshold selection, final evaluation, SHAP explanations, and saving a reusable prediction bundle.

Key finding	Evidence	Interpretation
Churn prevalence	1,869 of 7,043 (26.54%)	Class imbalance means accuracy alone is insufficient.
Contract association	$\chi^2=1,184.60$; $p=5.86\times 10^{-258}$; $V=0.410$	Contract type has a strong association with churn in this dataset.
Lifecycle pattern	Median tenure: churn 10 months; retained 38 months	Churned customers are concentrated at shorter tenure.
Final test performance	$F1\approx 0.600$; $\text{recall}\approx 0.655$; $\text{precision}\approx 0.553$	At threshold 0.45, the model detects about two-thirds of test churners, with false alerts as a trade-off.
Ranking metrics	$\text{ROC-AUC}=0.8189$; $\text{PR-AUC}=0.6141$	The model separates risk levels beyond a single cutoff; PR-AUC should be read against the imbalanced prevalence.

The final model is retained under the notebook’s F1-oriented comparison. Findings are associative and predictive, not causal: the analysis does not establish that changing a customer attribute will itself prevent churn.

1. Objective and Workflow

Objective: estimate customer churn probability, identify patterns associated with churn, compare classification models, and explain individual predictions.

Workflow: load and inspect data → assess quality → clean → explore → test associations → engineer features → stratified split → scale/encode → train baselines → cross-validation and tuning → SMOTENC on training data → compare candidates → select threshold from out-of-fold training predictions → evaluate once on held-out test data → SHAP explainability → save and verify deployment bundle.

2. Dataset and Data Quality

Item	Notebook observation	Decision
Raw shape	7,043 rows × 21 columns	Retain customer-level records.
customerID	Unique identifier	Drop from predictors; it does not represent customer behavior.
TotalCharges	Text/string field	Convert to numeric; invalid blanks become missing.
Missing values after conversion	11, all associated with tenure=0	Fill TotalCharges with 0 based on the zero-tenure context.
Cleaned data	7,043 × 20; zero missing values	Proceed with target encoding and analysis.
Target	No=5,174 (73.46%); Yes=1,869 (26.54%)	Encode No→0 and Yes→1.

Decision note: the notebook’s treatment of the 11 TotalCharges values is grounded in their zero-tenure records. This is a domain-based imputation choice, not a general rule for missing charges in other datasets.

3. Exploratory Data Analysis

3.1 Target balance

The churn class makes up 26.54% of records, while non-churn makes up 73.46%. A majority-class model could appear accurate while failing to identify many churners; recall, precision, F1, ROC-AUC and PR-AUC are therefore reported alongside accuracy.

3.2 Contract type

Contract	No churn	Churn	Observed churn rate
Month-to-month	2,220	1,655	42.71%
One year	1,307	166	11.27%
Two year	1,647	48	2.83%

Month-to-month customers have a much higher observed churn rate than customers on one- or two-year contracts. This is a prominent segmentation pattern and is statistically tested below. It should not be interpreted as proof that changing a contract would cause a specific reduction in churn.

3.3 Tenure and lifecycle

Tenure statistic	No churn	Churn
Mean	37.57 months	17.98 months
Median	38 months	10 months
First quartile	15 months	2 months
Third quartile	61 months	29 months
Maximum	72 months	72 months

The tenure distribution for churned customers is shifted toward shorter customer lifetimes. The median difference is substantial and supports treating early-lifecycle customers as a distinct risk segment for monitoring.

3.4 Charges, services and correlations

- Monthly charges: the notebook's boxplot and Mann–Whitney test indicate different charge distributions between churn groups; distributions overlap, so charges alone do not classify customers.
- Internet service and payment method: the EDA summary identifies Fiber optic service and Electronic check payment as segments with higher observed churn.
- Numeric correlation: tenure and TotalCharges are strongly positively related (approximately 0.83); MonthlyCharges and TotalCharges are also positively related (approximately 0.65). Correlation indicates overlapping information, not that a feature must automatically be removed.

Graph interpretation: the notebook uses target-count plots, demographic and contract comparisons, tenure and monthly-charge distributions, category-wise churn plots, and a numeric correlation heatmap. Visual differences are descriptive; the inferential tests and held-out model evaluation provide complementary evidence.

4. Statistical Testing

Relationship	Test	Statistic / p-value	Interpretation
Contract × Churn	Chi-square independence	$\chi^2=1,184.60$; $p=5.863 \times 10^{-258}$; Cramér's $V=0.410$	Strong evidence of association; effect-size measure indicates a substantial relationship in this sample.
Tenure × Churn	Mann–Whitney U	$U=2,515,538$; $p=2.420 \times 10^{-208}$	Tenure distributions differ strongly between churn groups.

MonthlyCharges × Churn	Mann–Whitney U	U=6,003,125.5; p=3.312×10 ⁻⁵⁴	Charge distributions differ significantly.
TotalCharges × Churn	Mann–Whitney U	U=3,381,224; p=5.685×10 ⁻⁸³	Cumulative-charge distributions differ significantly.

The notebook uses $\alpha=0.05$ for significance decisions. Very small p-values are evidence against the corresponding null hypotheses; they do not establish causality and do not, by themselves, measure predictive usefulness. Distribution plots, effect size and test-set model metrics add context.

5. Feature Engineering and Preprocessing

Feature / step	Implementation	Purpose / decision
AvgChargesPerMonth	TotalCharges / (tenure + 1)	Charge-related normalized signal; denominator avoids division by zero.
TenureGroup	0–1yr, 1–2yr, 2–4yr, 4–6yr	Interpretable lifecycle bands.
NumServices	Count of selected subscribed services	Represents breadth of service adoption.
Train/test split	80/20, random_state=42, stratified	5,634 train and 1,409 test; both have ~26.5% churn.
Numeric preprocessing	StandardScaler	Scale numeric predictors.
Categorical preprocessing	OneHotEncoder(handle_unknown='ignore')	Convert categories into model-ready indicators.
Feature space	6 numeric + 16 categorical source fields; 51 transformed features	Keep feature transformation consistent for training and inference.

The split is performed before balancing. SMOTENC is applied only to the training partition; the test set retains the original prevalence. This separation helps avoid synthetic minority examples contaminating final evaluation.

6. Baseline Models and Benchmarking

Baseline model	Accuracy	Precision	Recall	F1	ROC-AUC
Logistic Regression	0.8048	0.6678	0.5267	0.5889	0.8458
Decision Tree	0.7984	0.6347	0.5668	0.5989	0.8302
XGBoost	0.7984	0.6510	0.5187	0.5774	0.8408
Random Forest	0.7793	0.6105	0.4652	0.5281	0.8195

Baseline comparison shows metric-dependent trade-offs. Logistic Regression has the highest baseline ROC-AUC, while Decision Tree has the highest baseline F1. The notebook proceeds to tuning and imbalance analysis rather than choosing solely from accuracy or one ranking metric.

7. Imbalance Handling and Tuning

Training data stage	Non-churn	Churn	Share
Before SMOTENC	4,139	1,495	73.46% / 26.54%
After SMOTENC	4,139	4,139	50% / 50%

The balanced training matrix contains 8,278 rows and 51 features. The untouched test matrix contains 1,409 rows and 51 features.

Candidate on test set	Accuracy	Precision	Recall	F1	ROC-AUC	PR-AUC
Tuned XGBoost — balanced	0.7786	0.5764	0.6257	0.6000	0.8189	0.6141
Tuned Random Forest — balanced	0.7651	0.5497	0.6364	0.5898	0.8280	0.6190

Tuned XGBoost — imbalanced	0.8006	0.6515	0.5348	0.5874	0.8430	0.6515
Logistic Regression — baseline	0.8048	0.6678	0.5267	0.5889	0.8458	0.6577
Decision Tree — baseline	0.7984	0.6347	0.5668	0.5989	0.8302	—

For tuned XGBoost, balancing increased test recall from 0.5348 to 0.6257 and F1 from 0.5874 to 0.6000, while reducing precision, accuracy, ROC-AUC and PR-AUC. Thus, the notebook's selected balanced candidate reflects an F1/recall-oriented trade-off rather than the highest ranking scores.

Balanced tuned XGBoost parameters recorded in the notebook: `n_estimators=200`, `max_depth=8`, `learning_rate=0.1`, `subsample=0.9`, `colsample_bytree=0.8`, `min_child_weight=1`, `gamma=0`, `reg_alpha=0.1`, `reg_lambda=2`. Best cross-validated F1 was approximately 0.8369.

8. Threshold Selection and Final Test Evaluation

Threshold	OOB precision	OOB recall	OOB F1
0.40	0.8173	0.8908	0.8525
0.45	0.8313	0.8775	0.8538
0.50	0.8427	0.8608	0.8517
0.55	0.8564	0.8430	0.8496

The threshold 0.45 maximizes F1 among the evaluated out-of-fold training predictions shown. Lowering a threshold generally increases sensitivity but also raises false-positive alerts. The test set is not used to choose this threshold.

Final test metric at threshold 0.45	Value	Meaning
Accuracy	≈0.769	Overall correctness; not sufficient by itself under imbalance.
Precision	≈0.553	About 55% of customers flagged as churners were actual churners in this test set.
Recall	≈0.655	About 66% of actual churners were identified.
F1	≈0.600	Harmonic summary of precision and recall.
ROC-AUC	0.8189	Ranking separation across thresholds.
PR-AUC	0.6141	Precision–recall performance at the test prevalence.

Actual / predicted	Predicted No Churn	Predicted Churn
Actual No Churn	837 (TN)	198 (FP)
Actual Churn	129 (FN)	245 (TP)

There were 374 actual churners in the test set: 245 were detected and 129 were missed. Another 198 non-churners were flagged. These counts make the operational trade-off concrete and should inform any deployment threshold decision.

9. SHAP Explainability

The notebook creates a TreeExplainer for the tuned XGBoost estimator and calculates SHAP values for all 1,409 test rows across 51 transformed features. Dimension checks confirm that SHAP values and feature names align with the model input.

Prominent model signals described in the report	Interpretation
Contract_Month-to-month	Frequently pushes predictions toward higher churn risk.
OnlineSecurity_No	Absence of online-security service is influential in many predictions.
TechSupport_No	Absence of technical support is a recurring model signal.

PaymentMethod_Electronic check	Payment configuration contributes to prediction differences.
MonthlyCharges and tenure	Charge level and customer lifecycle contribute to risk differentiation.
Contract_Two year	Long-term contract status contributes toward lower-risk predictions in relevant cases.

Global SHAP summary and mean-absolute-importance plots describe how the fitted model uses features. They do not prove causal effects.

Individual example from the notebook: the first test customer is predicted No Churn with probability 0.31%, below the 0.45 threshold. A separately constructed example customer (12-month tenure, Fiber optic, month-to-month contract, electronic-check payment, and other specified attributes) is predicted Churn with probability 76.52%. Its waterfall plot decomposes the model output into feature contributions.

10. Deployment Artifacts and Verification

- The notebook saves the tuned model, scaler, encoder and feature names, and also creates a combined churn_pipeline.pkl deployment bundle.
- Reusable transformation and prediction functions apply the same scaling and encoding used during training, then compare predicted probability with the stored threshold.
- The notebook loads the saved bundle and verifies inference on the example customer; it reports a Churn prediction at 76.52% probability with threshold 0.45.

Deployment note: before production use, confirm that the saved model, preprocessing objects, feature ordering and threshold are versioned together. Monitor calibration, drift, and performance on newly observed outcomes.

11. Limitations and Reporting Notes

- The dataset is historical and observational; associations should not be treated as causal effects.
- Final performance is measured on one held-out test partition; ongoing monitoring on new data is necessary.
- The selected threshold reflects an F1 objective. Operational use should account for the relative costs of missed churners and false-positive alerts.
- SMOTENC creates synthetic training examples. Test isolation is maintained, but calibration and population drift still require monitoring.
- Tenure and TotalCharges share information; feature importance should be interpreted in that context.
- SHAP explains model behavior, not what would happen under an intervention.

Conclusion

The notebook documents a complete churn-analysis and prediction workflow. The strongest descriptive and inferential patterns concern contract type, tenure, service configuration, payment method and charge variables. The balanced tuned XGBoost model is retained as the final candidate under the project's F1/recall-oriented framework, with a 0.45 threshold selected from training out-of-fold predictions. On the untouched test set it achieves approximately 0.60 F1, 0.655 recall, 0.553 precision, 0.8189 ROC-AUC and 0.6141 PR-AUC. The resulting system can support risk prioritization and further retention analysis, provided its outputs are treated as predictive signals rather than causal prescriptions.

Appendix: Data Dictionary (selected fields)

Variable	Description
gender	Customer gender

SeniorCitizen	Senior-citizen indicator
Partner / Dependents	Household status indicators
tenure	Number of months with the company
InternetService	Internet service type
OnlineSecurity / TechSupport	Security and support service status
Contract	Contract term
PaymentMethod	Payment method
MonthlyCharges	Current monthly charge
TotalCharges	Cumulative charges
Churn	Target: whether customer churned

Graph-by-Graph Interpretation

This section explains the purpose, visible pattern, analytical meaning, and practical implication of the key figures in the notebook. Interpretations are descriptive unless a corresponding statistical test is reported.

1. Churn Class Distribution (Count Plot)

What it shows: The x-axis represents the churn label (No/Yes), and the y-axis shows the number of customers in each class.

Observed pattern: The No-Churn bar is much taller: 5,174 customers (73.46%) versus 1,869 churners (26.54%).

Interpretation: The target is imbalanced. A classifier can achieve apparently strong accuracy by favoring the majority class, while still missing churners.

Decision / implication: Use stratified splitting and evaluate recall, F1, ROC-AUC and PR-AUC in addition to accuracy. Consider imbalance handling only on the training data.

2. Churn by Gender

What it shows: The plot compares churn counts for female and male customers, split by churn status.

Observed pattern: The notebook's EDA does not identify gender as a standout churn segment; the group patterns are comparatively similar.

Interpretation: Gender appears less differentiating in the exploratory view than contract, tenure, or service/payment variables.

Decision / implication: Do not prioritize gender as a retention lever based on this plot. Confirm any small difference with appropriate group rates and fairness review.

3. Churn by Senior-Citizen Status

What it shows: The plot compares churn outcomes for senior-citizen and non-senior-citizen groups.

Observed pattern: The senior-citizen group is smaller, so raw bar heights should not be compared without accounting for group size. The notebook does not report a dedicated significance test for this plot.

Interpretation: A difference in counts is not necessarily a difference in churn rate; denominator-aware comparisons are needed.

Decision / implication: Use within-group churn percentages before drawing conclusions. Treat this plot as exploratory rather than causal.

4. Churn by Partner and Dependents

What it shows: These plots compare churn outcomes across household-status categories.

Observed pattern: The EDA uses these plots to inspect whether churn patterns vary by partner or dependent status. The notebook summary emphasizes contract and tenure more strongly than these demographic variables.

Interpretation: These variables may provide segmentation context, but raw counts can reflect the number of customers in each category.

Decision / implication: Compare churn rates within each category and avoid claiming household status causes churn.

5. Churn by Contract Type

What it shows: The x-axis shows month-to-month, one-year, and two-year contracts; bars separate churn and non-churn counts.

Observed pattern: Observed churn rates are 42.71% for month-to-month, 11.27% for one-year, and 2.83% for two-year contracts.

Interpretation: The plot shows a pronounced gradient: shorter commitment categories have higher observed churn. The chi-square test also finds an association ($\chi^2=1,184.60$, $p\approx 5.86\times 10^{-258}$; Cramér's $V\approx 0.410$).

Decision / implication: Contract type is a high-value segmentation variable. The association is not proof that changing a contract alone will cause churn to fall.

6. Tenure Distribution by Churn (Box/Distribution Plot)

What it shows: The x-axis separates churn groups and the y-axis shows customer tenure in months.

Observed pattern: Churned customers have a shorter tenure distribution: median 10 months versus 38 months among retained customers; the churn group's first quartile is 2 months.

Interpretation: Churn is concentrated earlier in the customer lifecycle. The Mann–Whitney test confirms a distribution difference ($p\approx 2.42\times 10^{-208}$).

Decision / implication: Early-tenure monitoring and onboarding analysis are reasonable next steps. The plot does not establish why early customers leave.

7. Monthly Charges by Churn (Box Plot)

What it shows: The x-axis represents churn status and the y-axis shows monthly charges.

Observed pattern: The notebook's EDA summary reports higher monthly charges among churners, although the distributions overlap.

Interpretation: Monthly charges distinguish the groups statistically (Mann–Whitney $p\approx 3.31\times 10^{-54}$), but the overlap means charge amount alone is not a reliable rule for identifying churn.

Decision / implication: Interpret price together with contract, service mix, tenure, and support features; statistical significance does not imply a large or causal effect.

8. Churn by Internet Service

What it shows: The bars compare churn and non-churn counts for DSL, Fiber optic, and customers with no internet service.

Observed pattern: The notebook's EDA summary highlights Fiber optic customers as a higher-churn segment.

Interpretation: Service type is associated with different observed churn patterns, but category sizes should be considered and the plot alone does not explain the mechanism.

Decision / implication: Review Fiber optic customer experience, pricing, and support as hypotheses to investigate—not as proven causes.

9. Churn by Payment Method

What it shows: The plot compares churn outcomes across payment-method categories.

Observed pattern: Electronic check is identified in the notebook summary as a higher-churn segment.

Interpretation: Payment method may act as a useful segmentation signal and may also correlate with other customer or billing characteristics.

Decision / implication: Validate with within-category churn rates and multivariable model explanations before proposing payment-process changes.

10. Numeric Correlation Heatmap

What it shows: Each cell shows the pairwise correlation between numeric variables; stronger positive or negative values are represented by stronger color intensity.

Observed pattern: Tenure and TotalCharges are strongly positively correlated (about 0.83); MonthlyCharges and TotalCharges are also positively correlated (about 0.65).

Interpretation: TotalCharges accumulates over time, so its relationship with tenure is expected. Correlated predictors can carry overlapping information.

Decision / implication: Keep correlation distinct from causation. Inspect model stability and feature importance jointly; do not remove variables solely because a correlation exists.

11. Baseline Model Comparison

What it shows: The chart compares model scores across Logistic Regression, Decision Tree, Random Forest, and XGBoost.

Observed pattern: Logistic Regression has the highest baseline ROC-AUC (0.8458); Decision Tree has the highest baseline F1 (0.5989). Rankings differ by metric.

Interpretation: There is no single winner independent of the project objective. Accuracy emphasizes overall correctness, while recall emphasizes catching churners.

Decision / implication: Choose the evaluation metric based on the cost of missed churners versus false alerts, then compare tuned models on the same held-out test set.

12. Confusion Matrix

What it shows: Rows represent actual classes and columns represent predicted classes. The notebook's baseline matrix is `[[937, 98], [177, 197]]`.

Observed pattern: The baseline correctly identifies 937 non-churners and 197 churners; it misses 177 churners and flags 98 non-churners as churn.

Interpretation: The matrix shows the concrete error types hidden by a single accuracy score. Baseline churn recall is about 52.7%.

Decision / implication: For the final thresholded model, the report separately records 245 true positives, 129 false negatives, 198 false positives, and 837 true negatives.

13. SHAP Summary / Feature Importance Plots

What it shows: SHAP summary plots show how transformed features contribute to predictions across customers; mean absolute SHAP values summarize average contribution magnitude.

Observed pattern: The notebook's interpretation calls out month-to-month contract, no online security, no tech support, electronic-check payment, monthly charges, and tenure as influential signals.

Interpretation: These plots explain the fitted model's behavior, not causal drivers. Correlated features can share or redistribute attribution.

Decision / implication: Use the global plot to understand overall model patterns and local waterfall plots to inspect individual cases. Validate stability before operational use.

14. Individual SHAP Waterfall Plot

What it shows: The waterfall plot starts from a baseline model output and shows feature contributions that move the selected customer's prediction up or down.

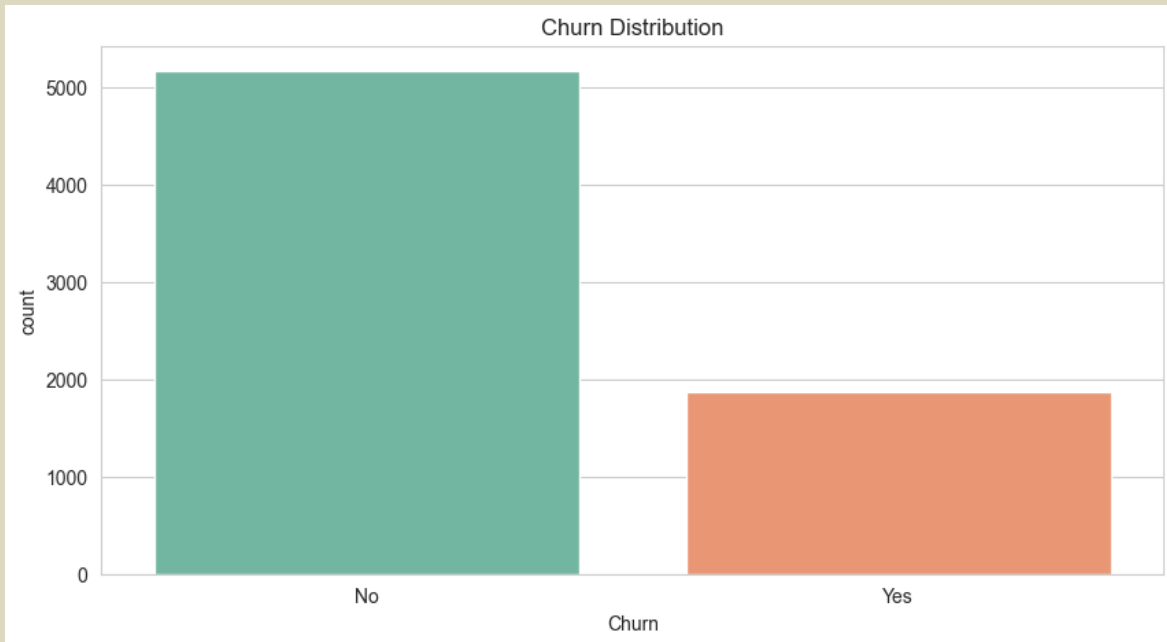
Observed pattern: For the notebook's example customer, the model predicts churn probability of about 76.52%, above the 0.45 decision threshold. The waterfall attributes this prediction across that customer's feature values.

Interpretation: The plot explains why the model scored this particular profile as it did; it does not say that changing one feature would necessarily change the real-world outcome by the displayed amount.

Decision / implication: Use it for case review and explanation, while retaining human oversight and checking the underlying input data.

Visual Analysis: Graphs with Interpretations

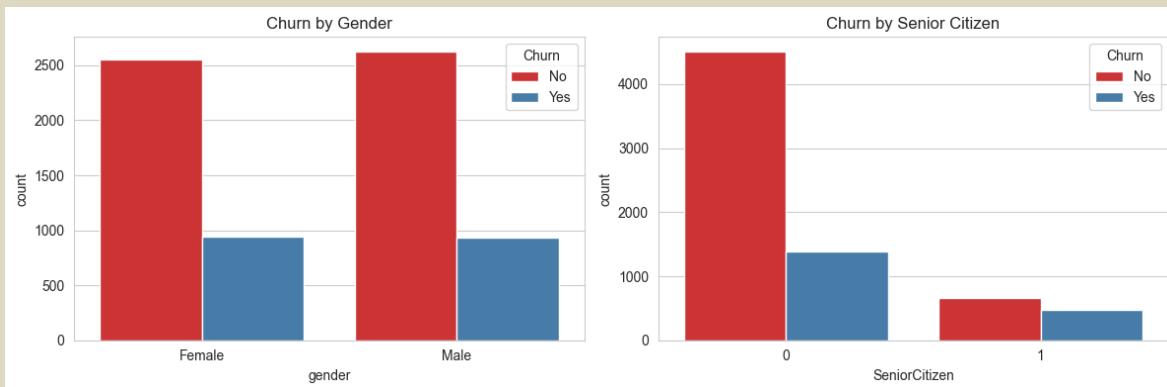
Customer Churn Distribution



Interpretation

The chart shows the class balance between customers who churned and those who stayed. Churn is the minority class (about 26.5%), so accuracy alone can hide missed churn cases. Model evaluation should therefore also consider recall, precision, F1 and PR-AUC.

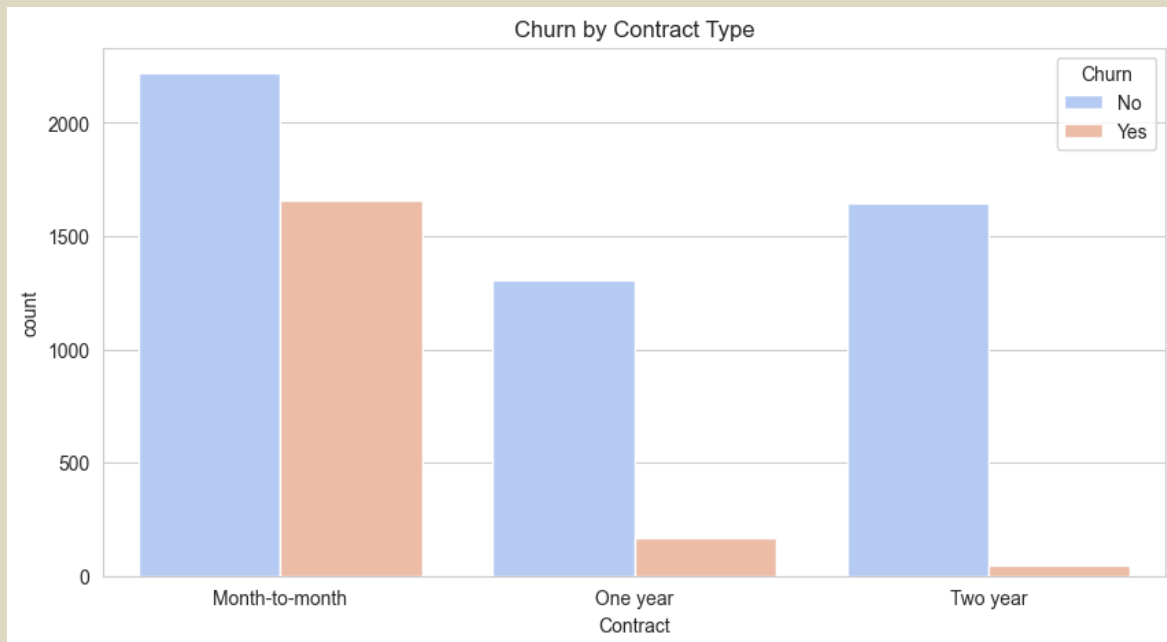
Churn by Gender



Interpretation

The gender-wise comparison shows broadly similar churn patterns across gender categories. Gender alone does not appear to separate churners strongly; retention analysis should focus on service, contract and tenure patterns alongside demographics.

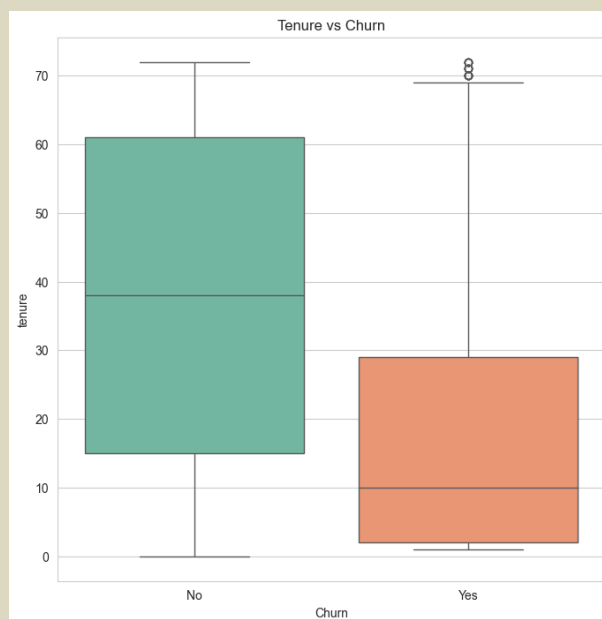
Churn by Contract Type



Interpretation

Month-to-month customers show substantially more churn than customers on one- or two-year contracts. This is an association, not proof that contract length itself causes retention; contract flexibility and customer tenure may also matter.

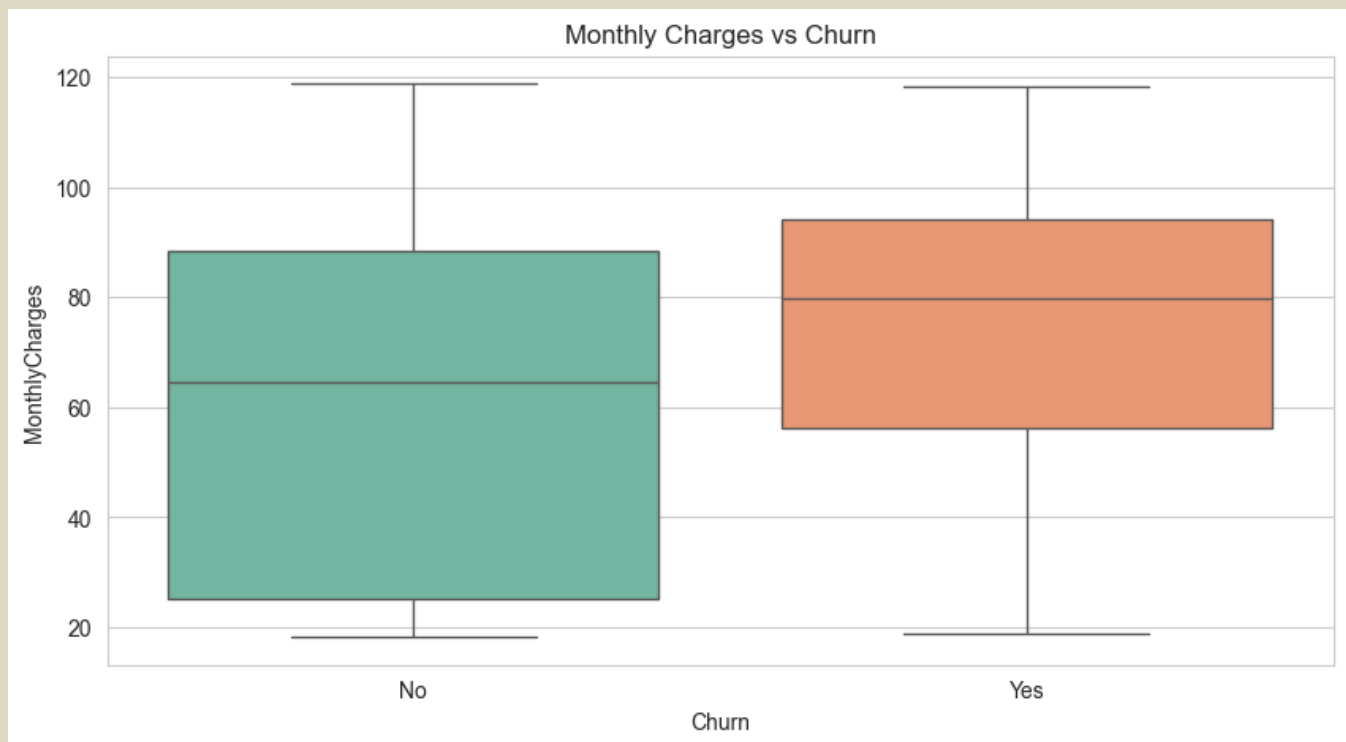
Tenure by Churn Status



Interpretation

The tenure distributions indicate that churners tend to have shorter customer tenure. This supports examining early customer-life stages and onboarding experience when designing retention analysis.

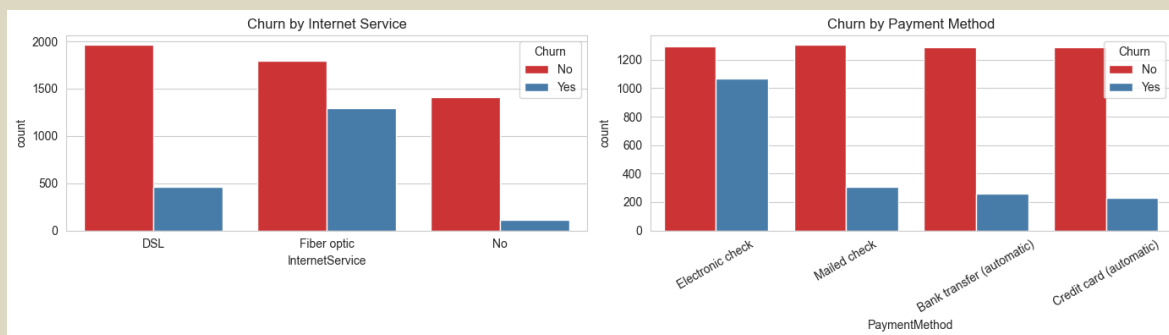
Monthly Charges by Churn Status



Interpretation

The boxplot compares monthly charges for churned and retained customers. Differences in the distributions suggest price level may be associated with churn, but charges should be interpreted together with the services customers receive.

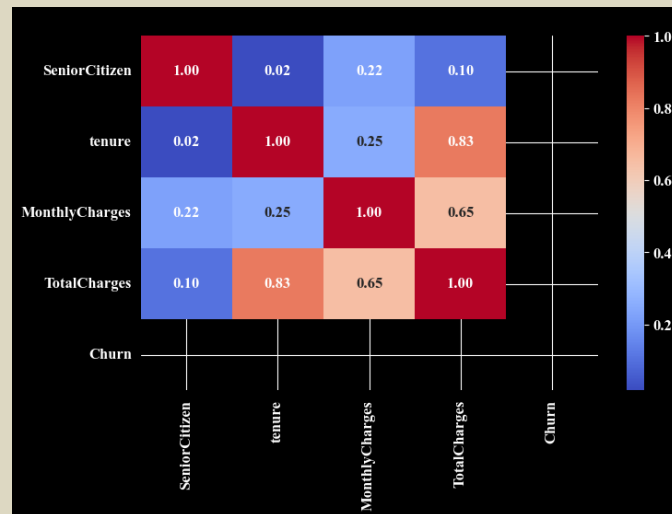
Churn by Internet Service and Payment Method



Interpretation

These category comparisons show that churn varies across service and payment categories. The patterns are useful for segmenting follow-up analysis, but category-level association should not be treated as a causal explanation.

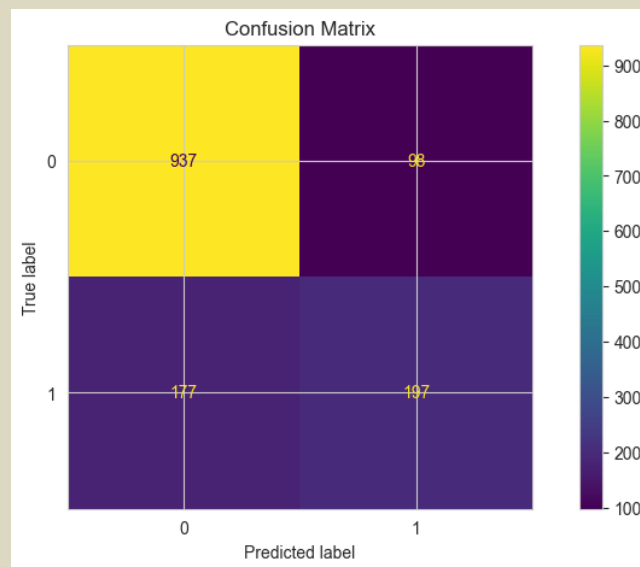
Numeric Feature Correlation Heatmap



Interpretation

The heatmap summarizes pairwise correlations among numeric variables. Strong relationships between tenure and cumulative charges are expected because total charges accumulate over time. Correlation does not establish causality and does not capture all categorical relationships.

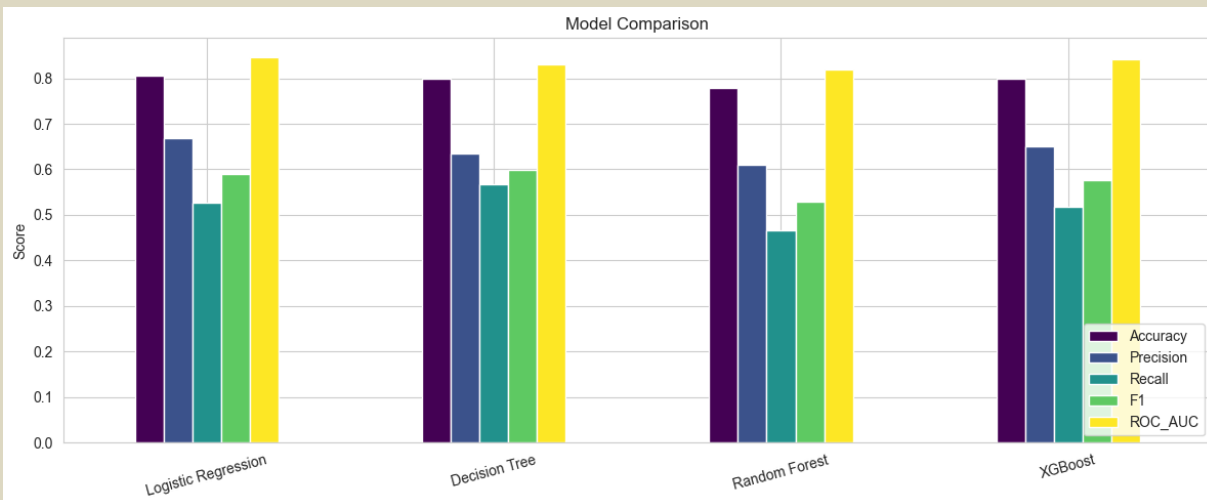
Confusion Matrix



Interpretation

The confusion matrix displays correct and incorrect predictions by class. False negatives are customers who churned but were predicted to stay; false positives are customers predicted to churn who did not. The acceptable balance depends on the cost of retention outreach versus missed churn.

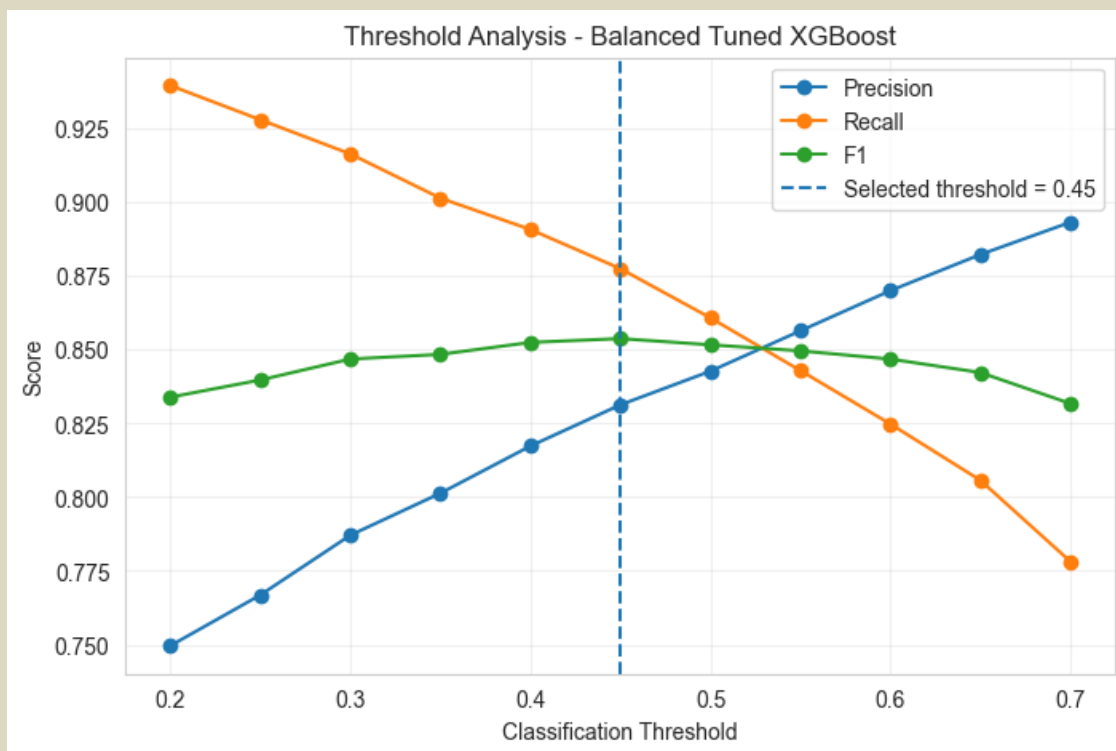
Baseline Model Comparison:



Interpretation:

The model comparison chart summarizes baseline evaluation metrics. Different metrics emphasize different outcomes, so the model with the highest accuracy is not automatically the most useful for churn detection. Recall, F1 and ROC-AUC should be considered together.

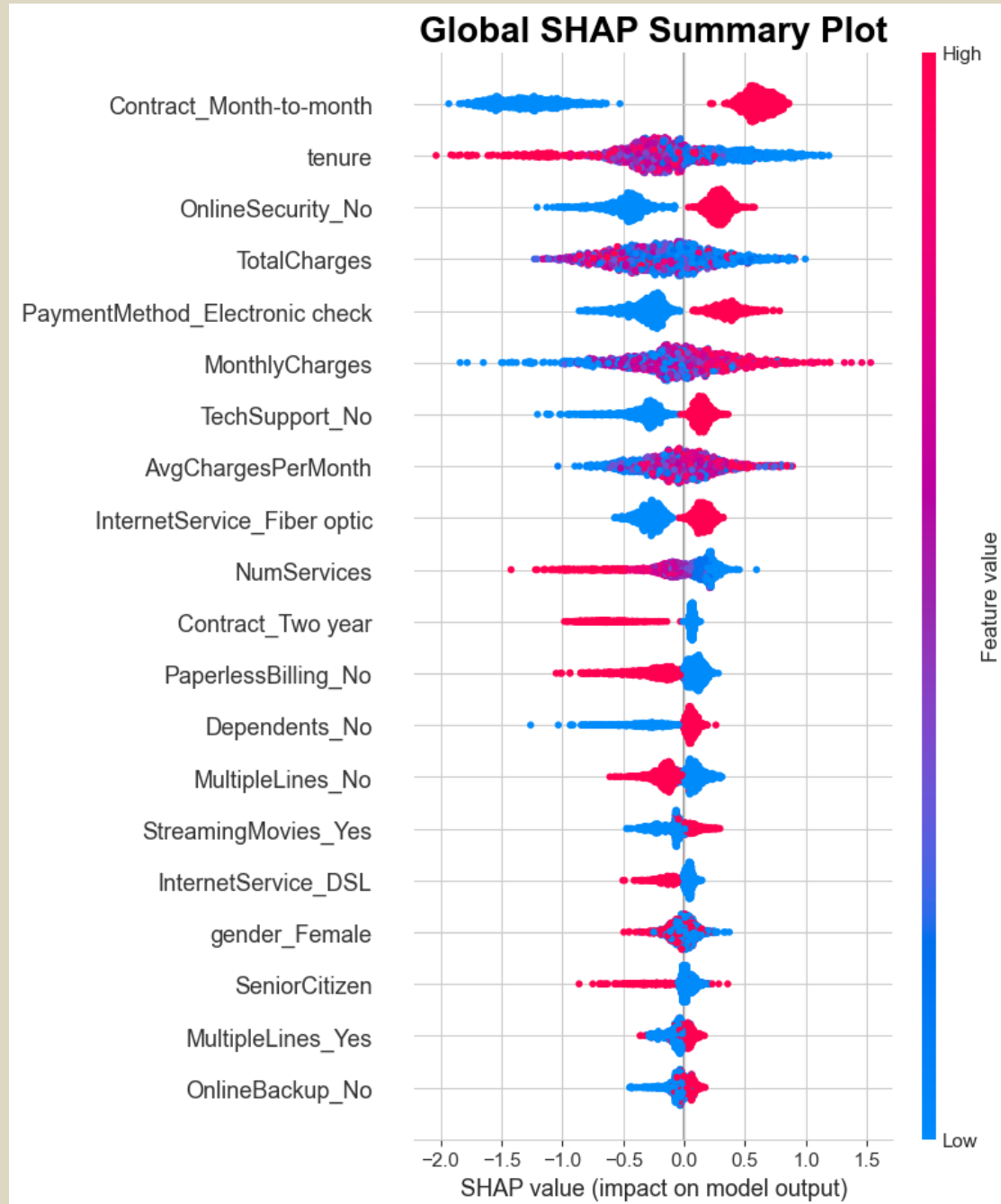
Threshold Performance:



Interpretation:

The threshold curve illustrates how classification performance changes as the probability cutoff changes. Lowering the threshold generally captures more potential churners while increasing false positives; threshold selection should reflect operational capacity and error costs.

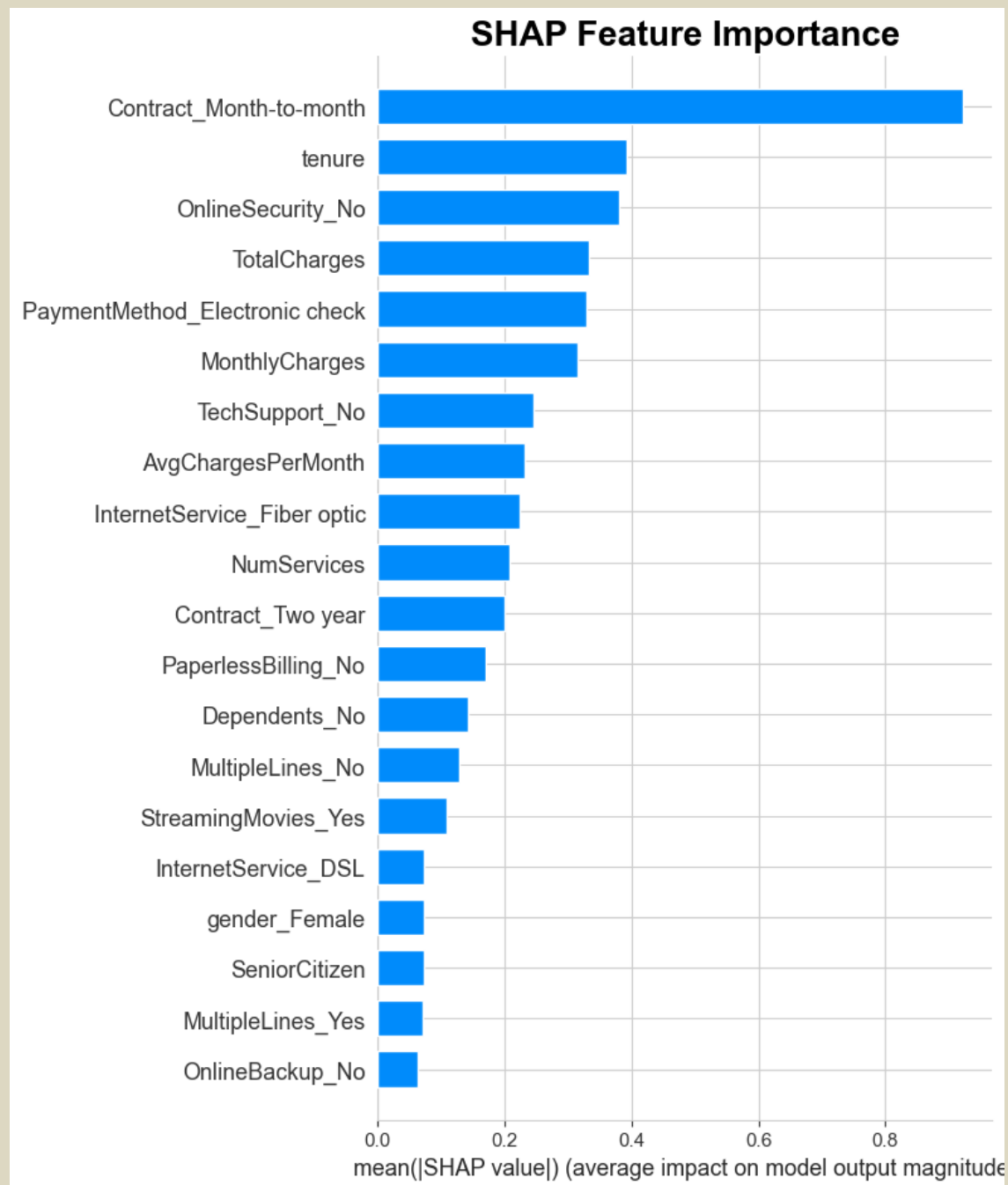
SHAP Feature Impact Summary:



Interpretation:

The SHAP summary plot shows which features contribute to model predictions and the direction of their contribution across observations. It explains model behavior, not causal drivers of churn.

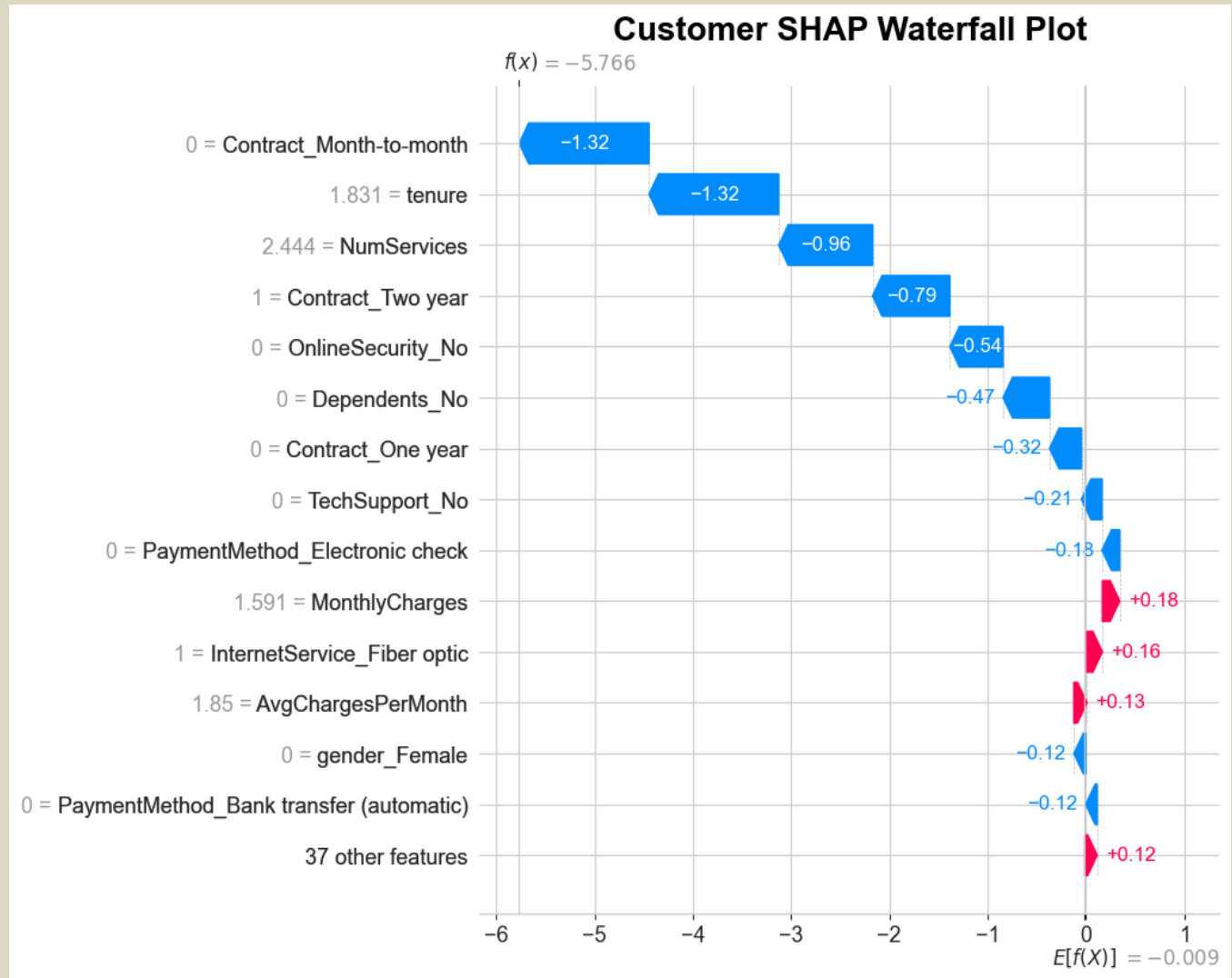
SHAP Feature Importance:



Interpretation :

This SHAP importance view aggregates feature contributions to show which inputs have the largest overall influence on predictions. Importance is model-specific and should be interpreted as predictive contribution rather than business causation.

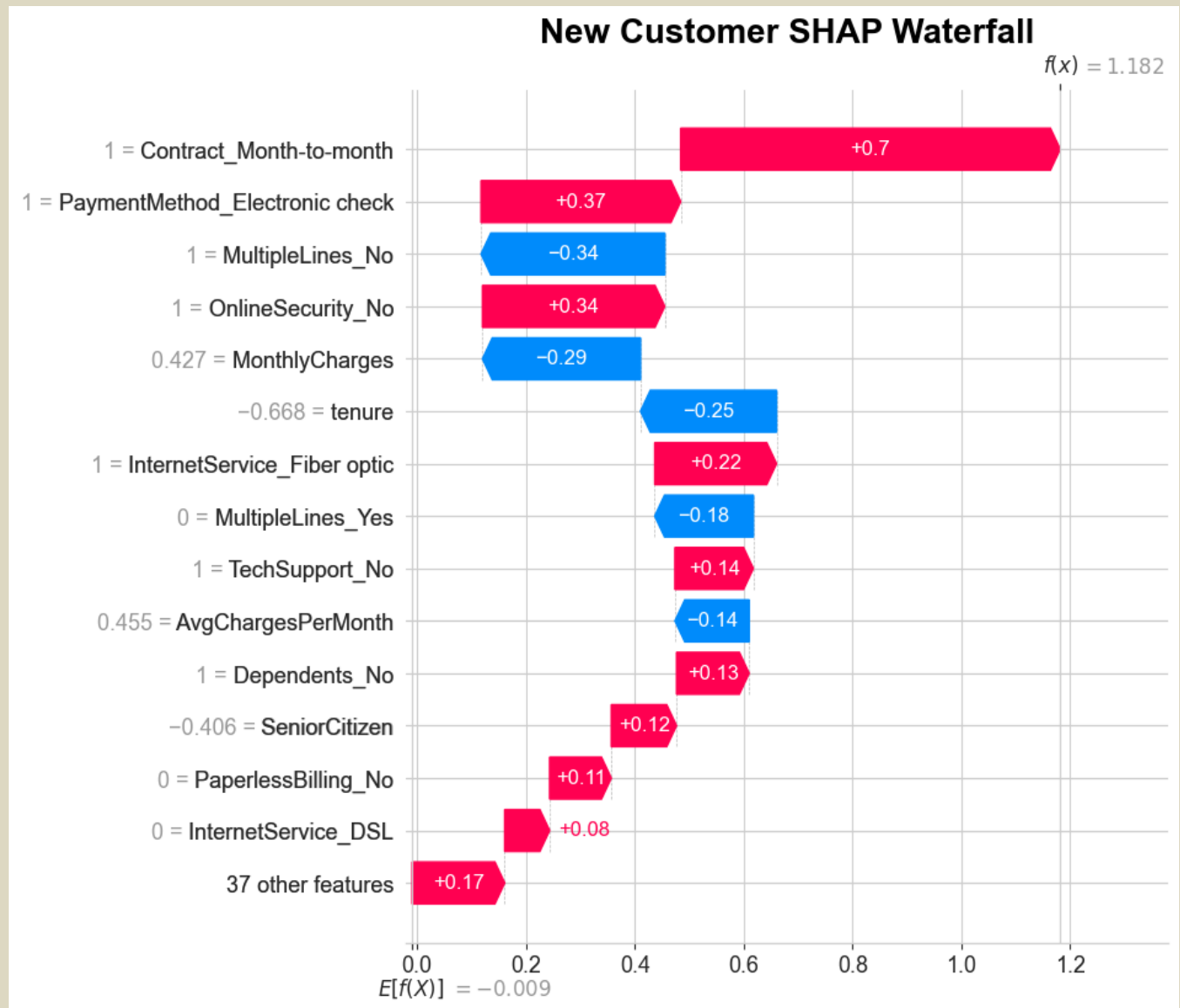
Individual SHAP Waterfall:



Interpretation:

The waterfall plot explains one customer's prediction by showing how feature contributions move the prediction from the baseline to the final score. This is an individual model explanation, not a general rule for all customers.

Individual Customer Explanation:



Interpretation

This visualization provides a customer-level explanation of the model output. The listed contributions describe why the model produced this particular prediction; they should be reviewed alongside the underlying customer data and model limitations.